

Kiera (Ngoc Thuy An) Tran

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[Google Scholar](#) | [LinkedIn](#)

Education

Ph.D. in Earth and Atmospheric Sciences

Anticipated 05/2027

Georgia Institute of Technology, Atlanta, GA

Advisor: Dr. Winnie (Wing Yin) Chu | [Group's Website](#)

B.S. in Earth and Atmospheric Sciences (Highest Honor)

May 2022

Georgia Institute of Technology, Atlanta, GA

GPA: 3.94/4.00

Publications

Tran, K., Chu, W., Yang, D., Dawson, E., Padman, L., Cordero, I., Tinto, K., Bell, R. (in prep).
Radar-derived basal melt rates across Ross Ice Shelf reveal regional variability controlled
by ocean circulation, tides, basal roughness, and subglacial discharge. *Sponsored by NASA
FINESST Fellowship.*

Field Experience

Patagonia Icefield Research Program, Torres de Paine, Chile

Nov 2023 – Jun 2024

Learned about geophysical methods and applications in glaciology and geomorphology

Self-led a ground-penetrating radar (GPR) survey

Observational Seismology Field Trip, South Carolina, USA

2022 – 2023

Deployed 86 passive SmartSolo seismometers and a broadband sensor to record the
sequences of earthquakes happening in South Carolina.

Geophysical Method Field Trip, Harvard Hill, California, USA

19 – 26 Mar. 2022

Applied geophysical instrumentation (magnetometer, GPR, BadElf Flex – GPS, and
seismic geophones) to detect active/inactive faults in the Harvard Hill of the Mojave
Desert, a part of the tectonically active San Andreas Fault.

Research Experience

Graduate Research Assistant, Georgia Institute of Technology, Atlanta, GA

Advanced approach for better attenuation loss calibration

2025 – present

Designed and tested computational methods for automated power fitting and loss
estimation to improve robustness of radar-based ice-sheet analysis, and improved
geophysical radar analysis through optimization.

An airborne study on West Antarctic ice-shelf basal channels

2022 – present

Applied machine learning and optimization techniques to airborne radar data for
detecting basal channel formation and tracking its evolution in West Antarctica.

Collaborator: Dr. Karen Alley - University of Manitoba, Winnipeg, Canada

Basal Melting Observations of Ross Ice Shelf (RIS) in Antarctica

2022 – 2026

Developed algorithms for estimating ice-shelf melt rates using radar attenuation and temperature modeling.

Modeling and Predict the impact of seawater intrusion Summer 2024

Simulated 100-year seawater intrusion impacts with ISSM modeling framework.

Collaborator: Dr. Frank Pattyn - Université libre de Bruxelles, Brussels, Belgium

Ice thickness measurement in Grey Glacier Summer 2023

Processed radar signals to estimate ice thickness and conducted field surveys for validation.

Collaborator: Dr. Camilo Rada - University of Magallanes, Punta Arenas, Chile

Bistatic Radar Development 2022 – 2023

Built a multi-transmitter/receiver radar system, including signal processing workflows for 3D subsurface temperature profiling

Undergraduate Research Assistant, Georgia Institute of Technology, Atlanta, GA

Forty-year of Ross Ice Shelf Change 2021 – 2022

Analyzed 40+ years of radar datasets of Ross Ice Shelf using **computational methods** for long-term stability observation & modeling.

Biological, ecological, and environmental flow applications Spring 2021

Conducted turbulence analysis of aquatic organism response using simulation and data-driven methods.

Conference Abstracts

- [13] **Tran K.**, Chu W., Yang D., Dawson, E., Padman L., Cordero I., Schroeder D., Bell, R. (2026). Airborne Radar Observations of Basal Melt Rates Linked to Cavity Ocean Circulation at Ross Ice Shelf, *Ocean Science Meeting*.
- [12] **Tran K.**, Chu W., Alley K. (2024). Airborne Radar Observations of West Antarctic Ice-Shelf Basal Channels, *AGU Fall Meeting*.
- [11] **Tran K.**, Chu W., Yang D., Padman L., Dawson E., Cordero I., Schroeder D. (2024). Understanding Ross Ice Shelf Stability from Radar Observed Basal Melt Rates, *AGU Fall Meeting - Oral Talk*.
- [10] Verboncoeur H., Hills B., Siegfried M., Holschuh N., Tarzona A., **Tran K.**, Schroeder D., Chu W. (2024). Internal and Basal Processes Driving Multi-Decadal Stress-Balance Changes in the Crary Ice Rise Region, West Antarctica, *AGU Fall Meeting*.
- [9] **Tran K.**, Chu W., Padman L., Schroeder D., Cordero I., Tarzona A. (2023). Integrating Basal Melt Rates from ROSETTA-Ice Airborne Sounding Radar Survey in Ross Ice Shelf, Antarctica, *AGU Fall Meeting*.
- [8] Tarzona A., Chu W., Verboncoeur H., Siegfried M., Schroeder D., Altaweel A., Amaro B., **Tran K.** (2023). Improved vertical Calibration of the Historical SPRI-NSF-TUD Airborne Radar Echo Sounding Ice thickness Measurements at Ross Ice Shelf, Antarctica, *AGU Fall Meeting*.

- [7] **Tran K.**, Chu W., Padman L., Schroeder D., Cordero I., Tarzona A. (2023). Integrating Basal Melt Rates of Ross Ice Shelf from ROSETTA-Ice Airborne Sounding Radar Survey and NASA/UCI Ice Sheet System Model, *WAIS Workshop*.
- [6] **Tran K.**, Chu W., Tarzona A., Schroeder D., Cordero I. (2022). Basal Melting of Ross Ice Shelf based on ROSETTA-Ice Radio-echo Sounding Observations, *AGU Fall Meeting*.
- [5] Tarzona A., Chu W., Verboncoeur H., Siegfried M., Schroeder D., Combs L., Prabu A., Altaweel A., **Tran K.** (2022). Geographical Repositioning Efforts and Vertical Calibration of Z-scopes from SPRI-NSF-TUD surveys at Ross Ice Shelf, Antarctica, *AGU Fall Meeting*.
- [4] **Tran K.**, Chu W., Tarzona A., Schroeder D., Cordero I. (2022). ROSETTA-Ice Observations on Basal Melting of Ross Ice Shelf using Airborne Radio-echo Sounding Radar, *WAIS Workshop*.
- [3] Tarzona A., Chu W., Verboncoeur H., Siegfried M., Schroeder D., Combs L., Altaweel A., Prabu A., **Tran K.** (2022). Archival airborne radio-echo sounding data geographical repositioning and calibration progress at Ross Ice Shelf, Antarctica, *WAIS Workshop*.
- [2] Tarzona A., Chu W., **Tran K.**, Teisberg T., Dawson E. (2021). Four-decades of Ross Ice Shelf Change: Part 2. Comparison using archival SPRI-NSF-TUD and modern ROSETTA-Ice and NASA/NSF IceBridge radio-echo sounding data, *AGU Fall Meeting*.
- [1] **Tran K.**, Chu W., Tarzona A., Teisberg T., Dawson E. (2021). Four-decades of Ross Ice Shelf Change: Part 1. Modern ice shelf basal conditions based on ROSETTA-Ice and NASA-/NSF IceBridge radio-echo sounding observations, *AGU Fall Meeting*.

Teaching & Mentoring Experience

Teaching

Polar STEAM

Research Scientist Fellow 2026 – 2027

Georgia Institute of Technology, Atlanta, GA

Facilitator (Teams for Tech)

GT Leadership Education and Development (LEAD) 2024 – 2025

Lecturer

GT 6000: Community, Mentorship, & Support Fall 2024

Guest Lecture

EAS 3610: Introduction to Geophysics, 'Heat Flow' 2023

Undergraduate/Graduate Teaching Assistant 2022 – 2023

EAS 1600: Introduction to Environmental Science

EAS 1601: Habitable Planet

EAS 2600: Earth Processes

Vien Ngo Temple, Loganville, GA 2023 – 2025

Language Instructor: Vietnamese

ABC-tutor, Atlanta, GA 2020 – 2022

K-12 & College Prep Private Tutor & Mentor

University of North Georgia
STEM Tutor

2018 – 2020

Mentoring

NASA Student Airborne Research Program

Summer 2026

Alissa Ferguson, Rice University

Catherine Pope, University of Vermont

Charlie Rosemond, Southern Utah University

Ella Boussy, University of Miami

Jackson Retelle, Embry-Riddle Aeronautical University

Morgan Fellows, Vermont State University Lyndon

Undergraduate Student Advising

Rowan Ray, GT Biology, 2024 – now

Daniel Park, GT Computer Sciences, 2024

Kieran Choi-Slattery, GT Aerospace Engineering, 2022 – 2023

Abdullah Altaweel, GT Computer Science, 2022 – 2023

Anh Nguyen, GT Industrial Engineering, 2022 – 2023

High School Student Advising

Huy Huynh, 2022 – 2023

Ben Huynh, 2021 – 2022 (*Now: Full-bright Undergraduate Student in Mechanical Engineering at UAH*)

Honors & Awards

NASA FINESST Fellowship (\$150,000)

2023 – 2026

West Antarctic Ice Sheet Workshop

Travel Grant (\$700)

2023

Georgia Institute of Technology, Atlanta, GA

EAS Travel Grant (\$1,600)

2025

SGA Fund (\$1450)

2024 – present

Center for Promoting Inclusion and Equity in the Sciences (\$500)

2024

CRIDC - Poster Competition Winner (\$1500)

Feb 8, 2024

President's Fellowship (\$22,000)

2022 – present

Quarter Century Award

2021 – 2022

Student of the Month

Oct 2021

Dean's List

2020 – 2022

University of North Georgia, Dahlonega, GA

Third Place College Division Nengajo Contest

2020

President's Honor Roll

2019 – 2020

President's List

2017 – 2019

AlaMATYC Math Tournament, Tuscaloosa, AL

<i>First Place Individual Calculus Competition (\$200)</i>	2017 & 2019
<i>First Place Team Competition</i>	2017

Academia experiences

Administration & Leadership

ACC Advocacy Days at Capitol Hill, Washington DC	2024 – 2025
<i>Advocated for increasing federal funding and more support and opportunities for international students</i>	
Georgia Tech Graduate Student Government Association	
<i>Worked with administrators from the President's Office to the department level to address concerns related to graduate students and build programs to support them, representing more than 30,000 on-campus and online graduate students</i>	
President	2024 – 2025
Executive Vice President	2023 – 2024
Vice President of Campus Services	2022 – 2023
Women in Geosciences, member	2022 – 2025
Graduate in Earth and Atmospheric Sciences, Secretary	2023 – 2024
Society of Exploration Geophysicists, Vice President	2022 – 2023

Academic Outreach

Georgia Tech, CRIDC	2023 – 2025
Georgia Tech, Geophysics Seminar, EAS department	2024
The Atlanta Children's Shelter, 'Invisible Ink & Chromatography'	Jul. 2023
Fort Valley State University, 'A way to study ice'	Mar. 2023
Georgia Tech, Graduate Student Symposium, EAS department	2023
Spelman College, Climate Sustainability Challenges & Opportunities	2023
Stone Mountain Middle School, 'Glaciers and Sea Level Rise'	10 Nov. 2022
Middle school visits, 'Glaciology'	Jun. 2022
Atlanta Science Festival, 'Earthquake Machine'	19 Mar. 2022
Georgia Tech, EXPLORE Science and Math!	2022

Guest Speaker

Du & Hoc Podcast, 'Save the Planet'	30 Oct. 2021
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Volunteer

Science Olympiad @ Georgia Tech	2025
Chattahoochee Riverkeepers	2018 – present
Wikipedia Editor	2020 – 2021

Relevant Courses

Machine Learning in Environmental Sciences	Advanced Data Analysis
Practical Computing & Math	Land Remote Sensing
Glaciers & Ice Sheet Dynamics	Sea Level Rise & Geocoastal Techniques

Introduction to Geophysics
Physical Volcanology

Geophysical Field Methods
Geodynamics

Technical skills

Coding languages:	Python, MATLAB
Libraries/Frameworks:	NumPy, Pandas, SciPy, scikit-learn
Tools:	Q-GIS, ArcGIS, Adobe Illustrator